ActInf Textbook Group ~ Cohort 5 ~ Session 18 (Chapter 8, Part 1) ~ 3/4/20

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all right welcome back everyone we're in
the first discussion for
chapter8 so does anyone want to begin
with like just
any thought or question on
eight any quote or or
part and we'll just see where it
goes
H yeah I was I'm was curious about the
role of
attractors um in
the yeah how how how the the role of
attractors in the um generative let me
let me try to find the
exact
quote
section yeah just basically the the Ro
of fixed point attractors in in the
generative
process anyone with a thought on
that all right there's a lot to say
about a tractor behavior in dynamical
systems overall and that's kind of where
we're
at one way to think about it is if there
wasn't some kind of attractor point or
set or limit cycle or something like
that that variable would like never
converge to aible
stationerity so if something's
oscillatory it could be thought of as
like converging to a cyclic attractor if
something is kind of like dampening like
a spring that's like a fix Point
attractor plus dampening or it could be

like just oscillating around a fix point
of tractor so it's
like dynamical systems are of certain
types analytically and they can be
converging and dampening converging
oscillating limit Cycles all these are
like different outcomes or possibilities
for dynamical systems but to kind of
pull back um a
step so chapter 8 follows on chapter 7
which shows the discrete time formalism
so in chapter 7 discrete time steps in
the simulation are explicitly modeled
like we have time 67 time 68
Etc chapter eight is going to show a
totally different complimentary
treatment of time for generative models
which is the continuous time formalism
in this chapter is about like unpacking
a lot of the differences from the
discret time and one of the key
differences is like a different take on
the relationship
between observations and hidden States
so in the um discreet time
setting and figure 4.3 is like the
Rosetta Stone with the top discrete time
model and the bottom continuous time so
in the partially observable marov
process setting the a matrix is kind of
going both ways between the hidden State
and the
observation here kind of in line with
the continuous time formulation being
like uh closer to some Physics
based uh
formula there's there's two core uh
equations there's observations

y same as observations here and then x
dot so the um derivative on hidden
States
so derivative on hidden States here and
basically each of these are mapped out
so there's like a kind of regularity of
observations or a function of
observations and a noise function and
then there's also an underlying
generator and this is kind of coming
from fris and at all's earlier work with
SPM where why are like all of the sensor
data for the neuroimaging like fmri or
EEG and then x dot is the rates of
change of the underlying neural activity
that's related to the observations that
are coming
in so that's kind of the that could
describe any G and any
F so then they by
8.3 get to a kind of
simple generative uh model where there's
a fix Point
attractor so here's the what it looks
like to have a fixed Point attractor so
this is basically when X is less than V
the set
point the rate of change is positive so
it goes up and then when X is higher
than the set point the rate of change is
negative so it goes down so it's kind of
like a first order
thermostat with a set
point encoded in the underlying Dynamics
like
this
any thoughts on that or does anyone like
want to add more pieces to this about

attractors or
anything yeah I think
um the the attractors become like
significantly more interesting whenever
we move into the the next uh sections
with like the the luck voltera Dynamics
and the lawen system just like trying to
map them to like other phenomena like I
really like the um the bird song example
I guess this is around 100 page
163 um where we're introduced to the
concept of generalized synchrony and so
the birds are basically like their
internal states are starting to
synchronize as they're like trading off
singing with one another and trying to
kind of match or continue the same song
so to speak and here we end up with this
kind of like like the states are
orbiting around in that lower left
figure two different I suppose
attracting points so it's like the state
is like orbiting between the two um but
then my my next question is I'm I'm not
sure exactly what those two orbits kind
of represent if they represent something
like one Bird versus another or if this
is like the state that one of the birds
as a single agent is like its beliefs
are moving around you know its
own um actions or or beliefs uh versus
like whenever it's not singing and
that's uh my next
thought
yeah that's a good question or or I mean
a good uh like progression
how first we have a simple fix Point
attractor so this is not even a moving

fixed point just a set fixed point then
kind of calling back to chapter five
there's a descending
setable fixed point so that was in the
the cross-section of the spinal cord
where there's the descending motor
prediction and then the proprioceptive
data coming in and then so it's kind of
it's the same exact mechanic as the
um set
point it just with the additional piece
that the set point is uh a descending
prediction
then
um another bump up in what the attractor
can be going to from stable fix Point
moving fix Point um now this gets into
the kind of like winnerless competition
or like generalized dynamical the
generalized sequential Dynamics these
are like iterative
processes and then first there's the um
this is kind of like a uh non-chaotic
winterless
competition with an ecological
fluctuation so lockable Terra arose from
the ecological Predator prey modeling
which is kind of what is referenced here
plant herbivore carnivore but also those
have been applied extensively to like
activity patterns of brain regions so if
you have or or Gene regulatory networks
like if you have multiple components
that have um s features such that when
they're low they can ramp up but then
when they're high they're like
self-limiting directly or indirectly
then they go up when they're low and

then they go down when they're high and
so they stay within like a bounded
interval but this is still has a kind of
a very
regular um movement so it's three
variables that are linked in a dynamical
equation p h and C in this case plant
herb carnivore and then here are like
two projections onto a side of the
cube and
then that gets taken into like one more
level to go from the non-chaotic
dynamical into the deterministic
chaos which is just kind of alluded to
and explored more in the
citation but
[Music]
then that takes us to the turn taking
Susan
so so I'm I'm wrestling
with um you the attractor States in
complexity science and and um chaos
theory and so I'm kind of wondering if
that's the steady state is you know is
the the
bounds in other words you you're going
to
move um you know it's
about how you're calculating the
movement and although I you I don't want
to might be better to put this
conversation um in the next after the
next chapter but um but in terms
of you calculating in it may be that
it's just a different nested model of
what is the current
state in other words in other words are
is this this state is going to look

different if it's if it's over
capacity it's out of a state of
resilience so I'm assuming that that's
actually another nested model and I
don't so I don't want to try to combine
the two or am I thinking about it
incorrectly yeah it's interesting like
explore it for specific system and then
maybe it'll be clear like which aspects
of
resilience are accommodated in the model
like as written like even the fixed
Point attractor I mean in a sense that's
a rudimentary form of
resilience because when it's pushed away
from the fix point it comes back to it
but there's multiple resilience Concepts
like one of them being that it returns
to where it was before another
resilience concept is that it stays
where it's adapted to so those are kind
of not even compatible definitions of
resilience which is like why when you
build a specific model then okay for
this variable when we say resilience
we're talking about the adaptivity like
definition versus like the resistance to
change version and then it's sometimes
really interesting how even some of the
simplest formalisms can
capture aspects of those
Dynamics but then
like you can't use the model to show
something outside of the model so there
if if this is the only one you have
you'll be able to have that kind of like
first order um fixed Point Behavior but
then it wouldn't be adequate to describe

some other kind of
behavior like this one this kind of
model will never show a moving set
point but this model would but it
wouldn't
show chaotic
Dynamics but then this one would so
head's trying all these different kinds
of models and just like like in the
textbook which is being shown here just
showing all these different features
okay then from the chaotic to the
chaotic with learning so it's just all
of these
different patterns and then it's kind of
like well for the phenomena for the
system that you're looking at which one
of those like levels or types of um
Dynamics feel
appropriate yeah and and and just to put
context around this this is just trying
to get give us the basics you know in
other
words not to throw in too much more
complexity I mean once we understand the
basics then we can you begin
adding more degrees of complexity I
guess Thomas Parr in his um book stream
gave a really
good kind of
timeline talking about how how how um
the earliest implementations were the
continuous State
space then people started thinking about
getting sequential Dynamics this is
exactly what we just saw with lock of
voltera then people wanted to look into
planning and sequential Dynamics with

like counterfactuals so then that
motivated more of the discrete space
discret Stace space discrete time model
to to get that kind of decision- making
and
planning um then there were hierarchical
models that were able to Nest either
discr or continuous or Hybrid
models um eventually by being
reintroduced introducing the continuous
State space as the kind of lower sensory
motor model and then having like a kind
of higher order discreet model and then
that was kind of explored in the folk
psychology work where we have continuous
time and state spaces like for sensory
and motor and then you have discreet
State spaces and time for more
sequential decision making and planning
Susan I think you just answered it I was
wanting a little more definition around
discrete space and continuous space but
anything
else
yes discrete State space is where the
options are finite and
discrete like one or two or three or
four or 0.1 or Point 2 or 3 but discrete
clickable values versus The Continuous
State space is when there's a Continuum
that's like differentiable like at all
points you could put a ruler and
determine the slope whereas if you only
have one two three four you can't really
like take a ruler and find the
derivative at each time step like you
can't you can't really have the
derivative in the same sense for a

staircase it's like a step function so
it's true The Continuous and the
discreete state spaces um we could be
talking about temperature like
temperature is a continuous variable
temperature is a discrete variable and
then one special variable for dynamical
modeling is how time is addressed and so
time as chapter 7 and eight Explorer
time can be explored as a discrete State
Space chapter 7 or time can be explored
as a continuous State space like chapter
eight
Andrew yeah just just like the way I
kind of think about
um the whole like discret space model on
the uh on the higher level of a
hierarchical model and then continues at
the lower um and I I don't know if this
stands up to anyone else but
um something like um I'm sitting here
I'm trying to figure out if my heartbeat
is too fast like yes or no to discrete
options I have to like pay attention to
my heart rate for some time but of
course like I can't just go with a
single time step I have to see like you
only know the rate by you know taking a
a second or a few seconds to figure out
what the rate is and so over you know
however many seconds that I need to
measure that which is multiple time
steps or a faster time scale I can then
make the discret decision if it's too
fast or not if that makes any kind of
sense so thus yeah and that's a perfect
um figure 8.6 that that um Daniel
pointed to is like we see how there are

multiple time steps in that continuous
level at the lower level U for every
single time Step At The Higher One um
and so yeah I think when thinking about
modeling like say an agent and trying to
include things like physiological uh
signals like heart rate and other things
like that there's this kind of faster
time scale going on and then the agent
at a higher level maybe a cognitive
level or some such thing is making like
more discret decisions or or inferences
based on what's going
on
yeah he anyone else have just any any
thought or
or question on eight we can look at some
questions we can
continue but the chapter's very
compact it just moves through those
sequence of
examples fix Point attractor fix Point
attractor moving point
attractor dynamical
attractor chaotic attractor with
learning
then it just says and you could put that
continuous model at the lower level of a
discreete Time
model and here's an example of what that
looks like like at the higher level is
which location to fix it upon and then
the continuous lower levels like the
Cade
movement and then there's a few settings
where that's been studied and that's the
whole chapter
J uh I have a little question about the

small question about this bird uh system
with the chaotic
attractors or chaotic Behavior so if I
remember right then chaotic
Behavior Uh the by definition it means
that you can't that you only have a
limited Horizon where you can predict
what is going to happen uh still these
birds seem to be able to
synchronize uh and is it if I look at
the figure
8.5 does it mean that they do not
synchronize their down to the right that
they even after learning they don't
synchronize perfectly they sometimes
lose each other but then they uh but
they stay quite close like they they
because if you can't predict what is
going to happen you probably can never
because you don't know the initial
conditions you know you never know
exactly where the other the other
individual is I guess but is that
correct
or yes so one thing to note um as uh as
I think it'll probably say
um singing from the same hym sheet so
just to be clear this model has
presented it's not like this is like a
generalized model of semantic
communication and turn-taking in
conversation this is like two birds with
a fixed innate song that they can kind
of in the second part of it do some
local tweaking but in the first part
they can't at all but they have the same
song template so they're not engaging in
more like open-ended um semantic

transmission it's kind of like like a
chant that's like supposed to be ongoing
and so the only options that they have
are either like listen and understand
that the other one is singing it or
notice that that has stopped and then
take action to to bring that song like
back into play that's kind of like the
situation that the bird song is modeling
um but then they show like in the paper
that when you have one bird with a
precise song and one with a very like
imprecise song that that the like that
younger imprecise one can like learn and
converge to the the other song then you
can kind of f fix that Bird's precision
and bring in a new plastic one and then
converge that one to it and so that got
into like the kind of inner generational
transfer again of like a templated song
with a little bit of Drift But yeah
what's Happening Here is like between
bouts of
singing There's this kind of tightening
of the joint Dynamics so this could be
like if this this would like two people
and they're pointing at each other
pointing and they're trying to they're
doing an improv game
where they're like trying to keep their
fingers as close to each other as
possible um so um if it the movements
were perfectly correlated then they
would be on like a onetoone
line like the x value of one would be
matched exact tracked perfectly by the x
value other that so and then if they
were totally uncorrelated it'd be like a

perfect circle which is say as wide is
as it is tall so like the correlation
would be zero there wouldn't be a
significant regression through that
cloud of points um so they start in the
same ballpark because they have a
template and then there's some fine
tuning in the learning process so that
their expectations tighten
up it's like and then the turn taking
part and then so about the
chaos and predictability so that's where
the um leop exponent comes into play
which is kind of like this variable that
describes how chaotic a system is
ranging from a system that's not chaotic
and the kind of metaphor there is like a
laminer flow so you have two planks of
wood on a laminer parallel flowing river
those two planks will stay near each
other at the same distance and then
there's other rivers where um planks
that are placed near each other
converge and then there's systems where
planks that are near each other diverge
and that's chaotic because you can't
necessarily know it's like you're coming
in on this one and then if you come in
just on this side you take another
attractor to the right but if you came
in a little bit more on the left then
it's spins out and so then as that
exponent gets more uh more chaotic it
gets harder and harder to predict points
um trajectories of points that start
initially next to each other so the
horizon predictability drops and so this
is kind of like it's like this one is

saying like we're on we're on the right
side of the turn taking and then it's
like the Precision is dropping and
that's kind of spiraling it out and then
eventually it spirals out and then it
flips to into the other attractor spins
in spins out and then eventually it
flips out
there just showing that those kinds of
of switching time Dynamics can be
modeled even with deterministic
equations
so there aren't even like this isn't
necessarily even
stochastic this is just the
deterministic chaos which was kind of
like that's one of the most interesting
and exciting parts from complexity
theory in chaos is like single pendulum
is deterministic and non-optic but then
the double pendulum is deterministic
still because it's just two single
pendulum but then it has the chaotic
Dynamics also connected to Wolf from
some
computational
irreducibility and about how like
there's things where even if they're
deterministic that it still can't be
determined like if the program's going
to do a certain thing or finish at
all yeah
Susan so I guess that would have a lot
to do with the policy
selection and is policy selection would
that would that be the same as you know
what model you
chose

to you know to you assess a
situation yeah like the singing Behavior
just on the first level the singing
Behavior can be understood as the policy
the motor policy of the bird okay
whether it SS or not like in a in that
binary system um but then a little bit
more cognitively the choice of which
framework to apply or which
interpretation to apply is like a mental
policy
decision so so where would affordances
come
in it's a good question in
um the of reference live stream 28 with
Mar sanit
Smith so this is kind of a different
this is another step by step so here's
static perception hidden State sensory
ambiguity to the observations D
prior now in the textbook they went from
that kind of static perception and then
went into the dynamic
perception but here there's kind of this
intervening step with a Precision
variable on a so this is like changing
the temperature or the prec the
Precision on
a which is to say from sharpening it so
that it looks more like just ones and
zeros and there's a less ambiguous
mapping between hidden States and
observations all the way in the low
Precision extreme a is just kind of like
blurred out so that observations have no
information on hidden state
so they're going the attention as
Precision here so now we have kind of

the discreete time figure 4.3 or figure
8.3 whatever it happens to be with these
little Precision guys on the bottom but
otherwise it's the
same and then let's say that this blue
is the sensory motor layer so that could
be the um spinal cross-section or that
could be the kind of gross motor
Behavior like
singing here's where the mental action
or the internal affordances the covert
action is coming into play where where
that affordances in that nested level
are modeled as basically Shifting the
Precision on the lower
level and then they had a third level of
policy about that second level policy
but everything past the blue sensory
motor layer are just higher mental
affordances oh very
cool yeah that was kind of a big moment
when the same or similar kind of um
formalism that was used for sensory
motor control and
affordances externally got brought in to
this kind of
internal attention as action setting by
by Lars and and others in
2021 and then they still like even in
the previous days continue
to like make this
more comprehensive and like just
bringing in more pieces and bringing in
like World Knowledge traditions in a
really interesting ways
too thank you yeah in discreet time I
don't know if there's any continuous
time versions of this but that would be

cool
too yeah Kristoff
yeah I just wanted to make a comment
that um I think it's interesting that
this formalism can actually describe
both um you know continuous time models
and discrete models because as we as we
move in in in the hierarchy of Concepts
then we finally arrive at at things that
are discrete but discrete phenomena in
some sense are not um um you know there
there's there's always this uh this sort
of uh problem of course graining from um
from underlying processes so there's the
generative process that gives rise to
the phenomena we are sampling that in
some way and our we're computationally
bound so we can't have perfect model of
the other system so we have to sample so
that gives rise to some kind of
discretization of uh of the phenomena
and temperature I think it's it's an
interesting example where you know we
can model it as a continuous variable
but it's not actually useful um to our
um uh for us temperature always has some
kind of discretization because we don't
care about I don't know six seven eight
decimal places right so so the fact that
you can build this hierarchy and this um
um uh discretization just kind of
naturally fits on you can fitted on top
of continuous models is actually U very
reassuring that this is a this is a
really good Paradigm that these two
things can be connected um kind of
directly without uh without bringing in
some um some other Theory to to say how

these how these systems would interact
um because that's just kind of what
naturally happens uh in in more complex
systems at the top level things do look
more and more binary uh when the
decision- making is made do they go left
or
right in the real world there is no
really left or right there's just
there's some kind of sensitive Direction
but um at my at my mental level that is
that that that might be a binary
Choice yeah very interesting thank you
um Kristoff or sorry um um your hand was
still raised on my computer but did Pros
did you raise your hand uh uh yes uh
actually I was uh Christoph actually
said what I was trying to say uh like I
was thinking like any kind of complex
system will have lot of continuous uh
items in the downstream but as you go up
there's like a decision making happening
and uh fin it's it's more binary but
it's like so the the the key learning
which I'm get getting over here is that
uh uh like there are various
possibilities in which you can mix and
match this kind of uh systems that you
want to have but there is a possibility
in which you can there's a flow and we
can use that for our own
purposes yeah that's very interesting
another kind of topic to connect this to
non-active um psychophysics is the just
noticeable difference concept so it's
like um the classic example is like we
have two objects that are going to get
put in one one in each of our hands and

then the task is which one is heavier so the objects have a continuous value value in reality or you could think of it as your muscles are exerting a continuous amount of tension to proprep and like a valuate the um mass of the object but it's a binary decision which one is heavier um so if the stimuli are exactly matched like if they're the same weight but there's no option for the the same weight you have to say which one you think is heavier but you'd expect that 50% of the time
like that it would be guessed which one is heavier and then it turns out that there's like this kind of sigmoidal curve where there's a high sensitivity when things are very close to each other to differentiate them but then if if it's like 5 pounds versus 15 pounds changing the 15 to a 16 it's not like it increases the decision-making accuracy by 10% because it's already at like 99.9% so that's kind of an interesting thing is like can connecting this the stimuli intensity to like potentially
the binarized
decisionmaking sigmoidal described decision making around that just noticeable difference um Point
Kristoff uh yeah there's also very interesting research on that on that subject um uh called qualia Quantum qualia they they're essentially testing a hypothesis that the the quality of experience depends on the history of experience so it's not that uh when you perform this kind of experiment say

asking people um uh which colors are uh
more similar and more different
depending on what the sequence of color
is that they've seen already their
responses um are likely to be different
so there are some people trying to test
theories that essentially bring in um
you know some kind of mathematics that
the kind of mathematics that's uh
previously been used in in quantum
theory to model uh model phenomena
almost like like a evolution of a of a
of a you know Quantum wave function so
that when when you're performing an
experiment it it really matters what
what what your context was and there's
some some really interesting results in
that um that space so uh there might be
you know it might be that there is sort
of like a reset time that's um that's
required but it might be that simply
that um we we we really perceive
deltas and that's what really matter so
if you give someone two very heavy
things and then to
ask them to compare two light things
they may just their brain might be used
to feeling the feeling of something
being heavy and not having enough
Precision not resetting enough not
having the Precision to perceive the
difference between the the the lighter
things so it's really really interesting
how um um how that
works
yes
yes and definitely like about detecting
change like the ey is expecting a

certain intensity of light and then it's
brighter than expected that induces like
a dilation or or a constriction however
it goes and then that recalibrates to a
new expected set point and then from
there again it's just about the
direction of like whether this is less
or more that's kind of the predictive
processing Paradigm which is like you
don't have to always be carrying through
like the temperature of the room if it
just happened to be that your implicit
set point was room temperature and then
you had that fix point of tractor all
you'd have to carry through would be
like plus or
minus within within reason and it
wouldn't need to be like
72.4 and do some sort of computation and
then store that you know you just have a
binary decision about which is action
dependent yeah so it seems to be that
like the bias seems we we always
continuously adjust the bias to to to
really see just the differences that
make the difference and I think there's
a really fantastic uh you know uh
example of this from Antiquity one of
the Greek philosophers um I forgot which
who it was but he was staring at the uh
at the river and um seeing people just
trying to uh I think rescue some animal
that just went into the river and and so
on and then and then he looked at the at
the ground after and he saw the ground
flowing so so even if it's not like a
physical um phenomenon like you know
expansion of the veins to pump more

blood or whatever um even just at the
level of the brain there is some kind of
processing that you can very easily see
yourself if you do it if you look at
something that flows like a water
flowing downwards on the TV and you look
at the wall you will see the wall
flowing because the brain adjusts to
that flow so that you could pick up the
differences within um you know the flow
not the not the flow itself you just
become oblivious to it um I wonder is
there any is there any work in um active
INF that would try to model something
like this because that that seems like a
really interesting
project I don't know specifically it
reminds me though also of like going on
a long walk or hiking and looking at the
ground a lot and then stopping and then
looking far away and they're kind of
like being like
zooming because there was even like a
calibration to the um the amount of
visual flow or just like an expectation
that like objects come closer or go
further or something like that and like
um another fun example like the
Exploratorium in San Francisco it's like
fun like Hands-On science was like a
pipe that had um a colder a hotter and a
room temperature component and then you
put like one hand on the colder and the
hotter part not burning or freezing but
just noticeably and then you put both on
the same section and then like one of
the hands fell the one that had been on
the hot felt like was cold and vice

versa but of course you're looking at it
and your hands are on the same piece of
metal but each hand was only able to
kind of like report that it was it was
getting contrasting opinions so felt
like as if you had flipped the hands but
they were both on the same
temperature and after images
visually some of these phenomena like
are more at the cellular level like some
visual after images and effects those
are like it's like some you know retinal
cells are exhausted and then like they
so it's like that that phenomena I mean
inquiry finds that that phenomena can be
explained by cell level findings whereas
something else like a ballerina that
rotates or a cube that's collapsed one
way or the other like those are not
going to Arise at the level of like the
single retinal firing
rate but they might arise at some
attentional
level
yes Andrew or anyone what what what is
quote what does the quote mean or how
does it start off the
chapter oh I just threw that in there in
reference to what Kristoff was
referencing uh I was I'm not sure if
it's herac Cletus who saw the um yeah
who saw the animals go into a river but
I know that this is a famous quote
referencing like yeah
there's another line like no one steps
through the same river twice like you're
you're
observing like you're observing behavior

and so everything is continuous but I
don't want to pull us too far away from
uh from the chapter with just saying
that but I mean I clearly to me it seems
to just be representing like continuous
Dynamics right continuous change over
time
so yeah totally
Pros no I was just uh trying to say a
little bit about the quote I think like
uh as far as I remember it's like you
cannot step on the same river twice
because uh after a moment the river has
changed you know like it is not the same
river it's like the concept like verbs
like it's like everything is changing
and it is not the same so
yeah just just a find yeah like so when
there's like a rock in the river and
then there's like a flow over it but the
flow is kind of like kept at a c like
there's a stable point for the flow
that's kind of like non-equilibrium
steady state because there's still pass
passing through a material dissipation
of a gradient but then if you were just
studying like the elevation of the river
at that point it would be a stable or it
would have like a little eing around a
small
point just on the kind of like practical
note um
pmdp the main python package for active
inference generative models is discrete
time only hence the PDP tutorials are
like all discreet
time whereas San jev's work in Python
accommodates continuous time models as

do the um mat lab earlier SPM work and
RX and fur and Julia but may maybe
someone has brought some continuous time
models in the PDP I just don't think
it's on like the main branch or or I'm
not sure if it's on any branch but it's
definitely in sanj's code SPM and in RX
fur the papers that are cited like at
the
end um we could double check but
just looking at um the the authorship in
the Years
these papers are probably almost all in
mat
lab we can double check
them but very
probably continuous time python Pros
um ask sanjie just in Discord just tag
sanj um is this wait Andrew is that a
continuous
time well I just sent for the general
documentation but yeah I mean all the
examples are
there yeah yeah this one's a discrete
time discrete time and discrete State
spaces actually
the this one has discreet and then the
continuous State spaces but it is it all
it is all in discreet
time but if you um in Discord maybe ask
sanj Andor like tag Magnus kudal and
sanj and you'll get to get answers okay
thank you yeah yeah um
like this this be fun collaborative
annotating for whoever wants to do so
like with with all of these examples and
more that we can
find um we could

annotate like time discreet continuous
either or like
hierarchical and then understand which
one of the code examples as we also
continue to add
them has these different features if
anyone wants to explore that we can
definitely do that but right now it's
just like the
links but but maybe look through them
and see which
um which ones have which
features
yeah
Christof uh yeah just I just checked my
sources and um the the story I was
referring to was um about Aristotle so
apparently he saw a horse stuck in a
flowing river and he fixedly watched the
rescue operation and he finally looked
away um and everything else appeared to
flow so he was uh it was the first um uh
recorded visual illusion uh in
history
wow horses kind of play a key role in
the visual perception because like the
first highspeed photograph was about the
debate whether the horses pick up all of
their feet when they run and then it
like they took a photo and it was
resolved like yes when they Sprint all
four of the feet are off the ground but
that was it was not known till that
happened well you know in the first
polish encyclopedia um written in in the
Polish language there was a very
interesting you know note about the
horse it just simply said said horse

what a horse is everybody can
see don't ask if you don't
know but you know it tells you something
about the you know about the
times
defitely there's been a lot of recent
work on the time perception and the flow
of time like Darius and and others and
some some upcoming live streams too have
kind of explored that that's definitely
a challenging topic because it's dealing
with something
experiential but people AR pursuing it
um whereas these examples like the eye
movement that's more
empirical um arguably this is
information like pupil arousal and and
icting this is is information that like
potentially every smartphone and webcam
collects all the
time I think this is you know one of the
interesting differences between machine
and human or machine and biological
vision is that um in biological
creatures we do have these mechanisms
and um of you know adjusting Precision
adjusting um resolution of picking up
salent uh features whereas machine
usually um and we have it a sort of at
the low level at the um um the level of
the eye that a lot of this processing
happens whereas in machine you just get
a big array of pixels and the neural
network is supposed to um figure that on
on its own uh from that input um and you
know I've been wondering for quite a
long time whether you know things things
things are the way they are for reasons

and we may not understand reasons uh why
things are the way they are but
evolution is not is not stupid it
produces things for uh um for for for
really for for really good reasons so I
I think that it might be that um the
fact that we have to sad to to to Really
look at something that we have this the
all these um all these limitations that
actually in in my view leads to more
robust uh represent ations um I don't
know how to test this hypothesis in in a
in a digital neural network yet but it
strikes me as as as highly plausible
that this limitation is necessary to
build um robust representations because
robust representations would be the only
thing that survives this kind of um
sampling
process and if you get too much detail
you you just get uh too much uh to it's
harder to extract information sine
information that's very interesting
whether there's like more fundamental um
like computational limitations that's
highly possible but like just when it
comes down to the fobia the center high
resolution area on the eye it's like
there are trade-offs just with the
density and the energy costs of having
high resolution in the center of the eye
and so like here's a visual Cade model
and then it's possible for this like
icade um agent to to differentiate these
different um visual patterns like is it
just line is it like parallel lines with
this diagonal thing or you know vertical
lines those can be quickly

differentiated like am I looking at a
Dorian or ionian pillar that could be
differentiated with like one to three
skillful is secades it's like am I
reading a scientific paper it's like
well is the lcav logo in the top left no
okay cic over here is there something
there no okay last check okay it's not a
paper that's whereas if um this were
going to be taken into like a sort of
classical machine learning pipeline then
let's just say this is like 20 by
20 well that's 400 pixels coming in so
you need like at the very least the
memory capacity for like that many
pixels to be coming in versus only nine
pixels with a 3X3 high
resolution and then that's like
the kind of Truth drop on our visual
experience being the generated not the
processed like no blind spot color in
the periphery resolution in the
periphery suppression of the Cades like
all those
features are the information that does
like give evidence in the moment to
moment that we're experiencing what we
generate
visually and then probably by extension
other senses too not like taking in a
whole whole scene and then distilling
the scene but having an underlying
generative model of the picture even if
it's our first time seeing that specific
one and then doing icting to fill in the
detail and then doing um all kinds of
other cognitive phenomena like
attentional modulation and kind of like

instantaneous overwriting to just like
make it normal make it feel
normal
yeah this this is uh I guess this will
be in one
month this has to do with the time
perception affect temporality
intention retain
pass the
sedimented retained embodied
trace and then the um unfolding kind of
time cone counterfactual
possibility prospective potention space
kind of just blurring
off in last minutes any like other
thoughts on
eight otherwise yeah it goes fast
basically that's eight then next week
earlier for Gort 5 we'll continue to
talk about eight maybe um Andrew if you
want to before then um or anyone else
like maybe we could look at one of the
papers in more detail like the is aade
paper with Thomas Parr just because I
mean we've kind of covered the
chapter in these conversations and we
can look at the bird song or or um some
other work with uh that's s of the
chapter then chapter nine that gets into
the empirical data basically up until
this point um all the discussion on
building the generative model the great
time in continuous time has been like
building it by
Fiat and then simul or making synthetic
data but then if you want to run it in
the other direction and take in
empirical data and then parameterize

your model for like determining within
or between group differences or doing
any other kind of testing like that's
where chapter nine comes into play and
then chapter 10 just summarizes it
up so it's so funny the book's like an
accordion like sometimes it seems like
there's a
ton other times it's just like barely
it's just like just quickly like
flipping through index cards of all
these different models because there's
so much detail to go into um with with
even the simpler
models so so just to confirm nine will
go will cover the um spatial
boundaries that when you said parameters
is you know we've talked about the the
time parameters and I'm trying to
um trying to Envision you how do you set
the spatial parameters that's what
Nine's going to
cover good good question parameters
could refer to spatial or temporal or
like cognitive any kind of parameter but
if we're going to be talking about data
coming from a spatialized system like a
room that has ensors in
locations then part of making a model to
deal with that data would be how do we
incorporate the spatiality into this
model so if our data have a spatial
component I mean if we don't take that
into account at the very least we've
left important information out about our
system that we
know worst case scenario we've come to
misleading conclusion because just kind

of like averaged out all the sensors but
they were in different locations you
know on the roof so then there's like a
really you know the roof is like a
triangle and then it was like there's a
temperature pattern that makes sense
when you know that some are on one side
and some are on another
side Okay the reason why time gets like
such a special
treatment whereas like space we could
model it continuously like we could have
a continuous GPS coordinate it with just
like as many levels of precision as we
wanted or we could discretize GPS and
just say okay we're only going to like
the first decimal point and then it's a
grid so the reason why that's less of a
big
deal is that the the time
concept gets taken into account every
single time we iterate the model
forward so how the model changes in time
is like obviously one of the key aspects
it is what makes it that
chemical so that's just a huge
decision that's why like in chapter
6 one of the big decisions that's
brought
up is like how is like should it be
handled should should it be discrete or
continuous because that it's a huge
difference whereas it's a smaller
difference if we
have um like temperature as a continuous
or a discrete value temperature could be
modeled continuous or
discrete whether we were thinking about

time discreete or
continuous so it's kind of like even
just with the thermometer in the room
like no action no nothing like this you
have a 2 by
two discrete temperature discrete time
continuous temperature
time off
diagonals like and it's kind of like
simple examples like that that give good
grounding as to why multiple models are
constructed because even if it was just
a air conditioner in a room you'd have
at least those four models at least in
principle and then you could explore
like which one have which ones have
different um which ones explain
different am proportions of the variance
of the
data which ones have different
computational resources and okay well
this one's more accurate but it's more
computational resources okay we'll run
that on the special one or we'll run the
cheaper analyses first and then if we
need to we'll rerun it for the final
version with the more
expensive you know those are just like
practical considerations that come into
play when you're working with real data
and systems
all right thank you see you all next
bye